Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

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Abstract

Agent swarm evaluation must distinguish increasing the total token budget, dividing a fixed ceiling, and changing coordination. We study these comparisons in 8,400 controlled trials on fresh arithmetic and graph-colouring tasks. At 4,096 completion tokens per worker, increasing independent width from one to four improves accuracy by 16.0 percentage points on both families while quadrupling the total ceiling. Splitting a fixed 4,096-token completion ceiling instead lowers graph accuracy from 56.0% to 18.5%; the arithmetic change is inconclusive. Primary comparisons do not establish an accuracy advantage over a sequential control matched on scheduled calls and nominal token ceiling. Ring communication provides no consistent gain. Eleven complementary scaling studies add 20,376 held-out trials across seven task families, including a fresh-task sixteen-worker confirmation that does not establish superiority over fixed-four execution. Candidate and request audits show that valid-answer recovery after initial failure depends on token allowance; they also reveal incomplete worker activation and 97 controlled-study attempts with unknown usage. Resource controls, output selection, and execution records thus qualify claims about agent scaling. The findings support evaluating specific allocation policies; equal realized compute advantages and general scaling laws remain unestablished from this work.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

> "All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).
>
> *Adapted to this study:* All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.

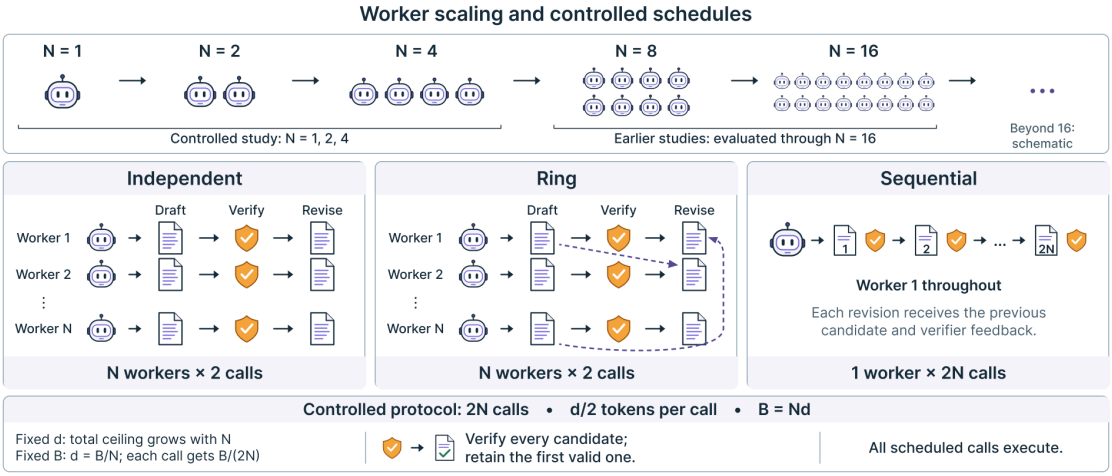


Figure 1 | Architectures and allocation: width N, per-worker completion allowance d, total ceiling $B = Nd$. All $2N$ calls execute with $d/2$ tokens per call. Controlled widths: 1, 2, 4; earlier studies: through 16. Larger widths are schematic.

1 Study and Findings

We evaluate worker width as a resource-allocation decision (Snell et al., 2024; Kim et al., 2026). **Budget rules.** Independent width $1 \to 4$ gains 16.0 pp on both families at fixed per-worker allowance, quadrupling the total ceiling; fixing the total lowers graph accuracy from 56.0% to 18.5% (arithmetic: -24.0 pp, inconclusive). PlanBench's independent four-worker effect reverses from -95.0 pp ($[-98.0, -91.5]$) to $+10.5$ pp ($[6.0, 15.5]$) as its ceiling doubles, with higher-ceiling debit ratio 4.06. Across 148 complementary wider cells (Figure 2), all use more recorded tokens: 75 score lower, 24 equal, 49 higher. These configurations share tasks and controls.

Worker participation. BigCodeBench Muse's $N = 16$ activation is 7.3%, 45.3% and 93.5% at ceilings 32,768, 65,536 and 131,072. At 32,768, independent widths 1/8/16 pass 40/64, 36/64 and 14/64. Activation's causal role remains unresolved.

Baseline choice. On fresh ZebraLogic tasks, transferred $4/8/16$ minus fixed four averages $+0.92$ pp over the two higher ceilings ($[-0.58, 2.50]$), establishing neither superiority nor equivalence.

Coordination. Sampling, aggregation and debate are distinct mechanisms (Brown et al., 2024; Wang et al., 2023; Du et al., 2023). Countdown's independent gains of 9.5–17.0 pp use 2.77–3.85 times recorded tokens; its pooled ring effect is -5.0 pp ($[-8.5, -1.75]$). Seven Muse graph zeros are operational failures, not necessarily reasoning errors. These ring results have narrower scope than shared-workspace communication (Park et al., 2026); execution audits complement failure taxonomies (Cemri et al., 2025).

Design and evidence. Figure 1 defines the controlled schedules and widths. Sources are BigCodeBench, MBPP/EvalPlus, ZebraLogic, BBEH and PlanBench (Zhuo et al., 2025; Austin et al., 2021; Liu et al., 2023; Lin et al., 2025; Kazemi et al., 2025; Valmeekam et al., 2023); custom graph/Countdown tasks use deterministic checks, with graph generation inspired by Reasoning Gym (Stojanovski et al., 2025). Complementary trials number 26,648: 6,272 calibration and 20,376 evaluation. Task bootstrap retains configurations and repetitions, stratified where applicable (Efron, 1979; Field and Welsh, 2007); 95% CIs lack cross-study multiplicity correction.

These results motivate evaluating allocation policies with recorded usage, realized participation and competitive controls. Ongoing work tests policy transfer across resource divisions; general scaling laws remain unestablished.
https://github.com/PluvioXO/agent-swarms-reproducibility.

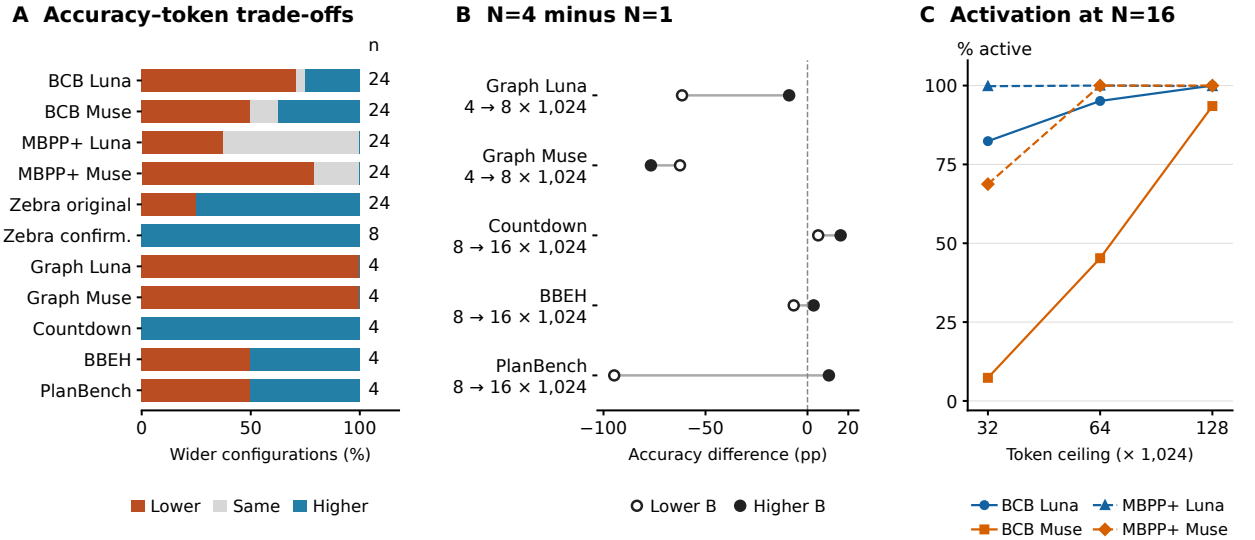


Figure 2 | Complementary studies: (A) accuracy of 148 wider cells versus matched-ceiling $N = 1$; (B) four-worker effects at two ceilings, pooled over revision conditions; (C) activation at $N = 16$.

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